

Lecture - notes

SLAM- basically no optimal solution
depends on the assumptions made

Lecture 3

(Relative Orientation, Fundamental and Essential Matrix)

- camera pair can be arbitrarily placed
 - 6 parameters are impossible to determine if global system is not known
 - We don't know the scale with two parameters
 - slide 9
 - if you know how big things are in the world (like ml)
 - photogrammetric model
 - don't know size
 - and global model/rf
 - Triple spat product must be zero -> to form plane (coplanarity)
 - skew symmetric matrix is the negative of its transpose
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2nd: Fundamental & Essential Matrix Direct Solution

- You have ≥ 9 points fixed in the plane (on the camera image)
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- 23: second nx9: zeros in lower half

Lecture 13

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- No scale ambiguity - we expect it to be known or same camera

Lecture - 15: ICP 3

when using symmetric point to plan: since serial input, transform the normal vectors

- Apply the Kernel by adjusting the weight
- Application is always trade-off

- Non-rigid registration e.g. with plants make the transformation even more complex